

Exploring Overreliance on AI Tools in English for Specific Purposes Courses: Challenges and Implications for Learning and Academic Integrity

Nataliia Avsheniuk 

Foreign Systems of Pedagogical and Adult Education Department
Ivan Ziaziun Institute of Pedagogical and Adult Education of the NAES of Ukraine, Kyiv, Ukraine

Nataliia Seminikhyna 

Department of Foreign Languages, Faculty of Economics
Taras Shevchenko National University of Kyiv, Ukraine
Corresponding author: nseminikhyna@gmail.com

Larysa Ruban 

Department of Foreign Languages, Faculty of Economics
Taras Shevchenko National University of Kyiv, Ukraine

Yuliia Sviatiuk 

Department of Foreign Languages, Faculty of Economics
Taras Shevchenko National University of Kyiv, Ukraine

Received: 02/09/2025

Accepted: 04/03/2025

Published: 04/25/2025

Abstract

The growing integration of artificial intelligence (AI) tools in education has generated considerable interest in their potential effects on students' learning outcomes, especially within the context of English for Specific Purposes (ESP) courses. This study explores how ESP teachers perceive and manage the impact of AI tools on student learning, particularly in writing, vocabulary development, reading comprehension, and problem-solving. The study focuses on the following primary research question: How do teachers perceive and navigate the impact of the tools on student learning outcomes and academic integrity in ESP courses? The research study employed a qualitative approach to investigate the impact of tools on English teaching and learning. The data collection involved 16 ESP university teachers from three faculties. A structured questionnaire was developed based on a comprehensive review of relevant literature with three main dimensions: (1) academic integrity concerns, (2) overreliance on AI that hinders learning outcomes, and (3) challenges of integrating AI into ESP teaching and learning. The study findings reveal that students' overreliance on artificial intelligence tools negatively affects language skill development, engagement, academic integrity, contextual understanding, and collaborative learning. One of the primary recommendations was to determine the specific boundaries for using AI, and to view AI as an academic aid, not a replacement.

Keywords: Academic integrity, AI tools, English for Specific Purposes, language learning and teaching, Ukrainian university, university teachers

How to cite: Avsheniuk, N., Seminikhyna, N., Ruban, L., & Sviatiuk, Y. (2025). Exploring Overreliance on AI Tools in English for Specific Purposes Courses: Challenges and Implications for Learning and Academic Integrity. *Arab World English Journal (AWEJ) Special Issue on Artificial Intelligence*: 3- 20. <https://dx.doi.org/10.24093/awej/AI.1>

Introduction

The growing integration of AI tools in education has generated considerable interest in their potential effects on learning outcomes, especially within the context of English for Specific Purposes (ESP) courses. AI technologies have a significant opportunity to address the need for improving instruction through real-time assessment, adaptive feedback, and personalized learning (Kem, 2022). This results in a more engaging and personalized language teaching experience, catering to a broader spectrum of student needs and learning styles. However, there are apprehensions about their learning outcomes relevance where their application is visible, such as in ESP- classes. Though such tools can be used to shortcut writing, vocabulary, and reading processes, over-reliance on them can short-circuit key cognitive processing skills needed for effective language learning and academic integrity.

The growing availability of digital tools and resources and academic performance pressures have made these environments complex to maintain integrity. The AI reliance creates a passive learning atmosphere, preventing students from actively interacting with course material and improving language acquisition. Often, outputs generated by AIs do not provide the kind of nuanced vocabulary and contextually relevant insights that specialized fields require, leading to inaccuracy or just a shallow understanding of the topic at hand. Relying on it denies students an opportunity to immerse themselves in the content, skipping over essential steps such as drafting, revising and critically analyzing – critical for establishing language proficiency and subject matter mastery. Although they may contribute to ESP learning processes, the gradual growth of pen and paper skills demise seems to be a huge concern for developing critical thinking skills and appropriate language use in a professional and academic environment. Therefore, there must be a regulation to control the use of such AI tools. There is a large gap in the empirical research regarding how AI shapes the development of language skills and ethical considerations leading to students' over-reliance on AI in their ESP learning experiences.

The study analyses how AI affects students' progress in ESP courses, especially in writing, vocabulary, reading and problem-solving tasks. It aims to determine whether over-reliance on AI tools hinders or facilitates learning and discuss ways to minimize dependence on AI while reaping its benefits.

This research aims to explore the perceptions of teachers about the impact of AI tools on student learning outcomes in ESP courses on tasks involving writing, vocabulary, reading and problem-solving, with the following core research questions: What strategies do teachers find effective in aligning AI with authentic learning experiences and maintaining the academic integrity of ESP classes? The findings highlight the challenges faced, as well as recommendations for educators on balancing the use of AI with authentic learning experiences and academic integrity.

Literature review

In education, the role of technology is becoming increasingly essential and is seen as a significant pathway for innovation within this sector. For genuine transformation in education,

fostering an environment where students can develop critical thinking through ethical principles and dialogue is crucial. There are ongoing concerns about its extensive application and possible adverse impacts on cognitive skills, especially in academic writing (Liu et al., 2023). Since research and education are deeply embedded in evidence-based inquiry, students must develop critical thinking, decision-making, and analytical abilities to effectively assess and interpret the quality of information (Hanim et al., 2020). Students must develop cognitive skills to synthesize, evaluate, and formulate well-structured arguments (Kaeppel, 2021).

Plagiarism is generally defined as using the ideas or words of another author inappropriately, representing them as one's own by not using a proper attribution (Park, 2003). It is also described as "literary theft" involving stealing words or ideas beyond common knowledge (Park, 2003, p. 472). However, the very basis of this concept faces challenges as to whether plagiarism is indeed theft or whether it can even be committed unintentionally, as discussed by Meuschke and Gipp (2013), whereby plagiarism can involve self-plagiarism or arise through improper citation, even though the original author may have had their permission granted. According to Maurer, Kappe, and Zaka (2006), there are two types of plagiarism: intentional and unintentional; the first "is the result of a deliberate effort to mimic someone else work without proper attribution" and to mislead, while the second "comes from not understanding the concept of plagiarism and citation" (Mahmood, 2009).

In ESP courses, the overexploitation of AI technologies prevents students from developing critical thinking and professional language skills. We all have realized that generative AI has brought opportunities and challenges in recent years. When used responsibly, AI can be a powerful tool to enhance the quality of education, but it also poses opportunities and challenges, particularly in academic writing (Kasneci et al., 2023).

AI-driven technologies can lend their hand to education in various ways, including but not limited to developing adaptive learning environments that fit each student's needs and preferences. Building an adaptive learning environment aims to improve the quality of learning and student performance in the educational process (Marengo et al., 2023). Integrating AI in higher education institutions (HEIs) has created specific challenges in preserving academic integrity. With the rise of AI tools in the educational domain, the possibility of their misuse as a vehicle for academic dishonesty has expanded, especially within the current virtual, remote learning environment. The very institutions that profess to uphold integrity have often been considered sources for this type of dishonest activity. Over the past two decades, extensive research has explored the implications of academic dishonesty, highlighting its negative impact on individuals' academic and professional trajectories (Akbar & Thomson, 2016; Eaton & Edino, 2018; Stoesz & Eaton, 2022). This growing body of literature underscores the importance of addressing issues such as plagiarism, data falsification, and other forms of misconduct to safeguard the credibility of academic systems (Guerrero-Dib et al., 2020; McCabe et al., 2001; Uzun & Kilis, 2020).

Over the past decade, AI has profoundly transformed English language teaching (ELT) through various innovative applications and tools that benefit learners and educators. AI tools are

now widely utilized in language teaching and learning, garnering significant attention from researchers aiming to explore the pedagogical implications of integrating AI into the classroom. In the context of, in particular, there is research on implementing AI-powered tools such as AI-based testing to improve ESP students' knowledge of tenses (Ningsih, 2023), chatbot to enhance students' English vocabulary (Qasem et al., 2023) and to improve Business English grammar and vocabulary (Kovačić & Bubaš, 2023).

Discussions among students about limitations and bias in AI and its appropriate use are shown (Chaudhry et al., 2022). This allows learners to improve their critical thinking and decision-making skills by promoting informed decision-making concerning AI. This can transform language education through AI tools such as ChatGPT that provide customized, flexible, and engaging learning experiences that closely resemble authentic language use and encourage collaborative learning (Abdelghani et al., 2022). Analyzing task-based ESP case studies with AI transforms each generation of students, helping them gain the necessary skills and domain-specific language structures. As one of the most acknowledged academic learning outcomes, critical thinking involves reasoning, inference, evaluation, and creative problem-solving (Facione, 2013). AI-powered tools can assist in critical thinking by prompting students to back up their claims with evidence, challenging students to consider opposing viewpoints and guiding them to explore multiple reasoning paths. AI's aptitude for mimicking dialogue and forceful counterpoints improves students' assessment of information critically rather than passively accepting it (Trang, 2021; Young, 2003). AI may provide significant benefits in creating critical thinking. However, scholars emphasize the need to regulate its limitations and ethical implications. Avsheniuk et al. (2024) argue for a balanced approach to incorporating AI into education outlets. The study highlights the importance of ensuring that AI enhances, not replaces, human interaction. They see AI as a potential for cognitive skills development and an opportunity to assist in acquiring increasingly important language skills. However, they also believe that there should not be an overreliance on technology to the detriment of inquiry-driven by humans. These challenges, from AI bias to misinformation and dependency, are ethical challenges we must resolve. Researchers are calling on educators to steer students towards actively responsible use of AI, positioning AI as an inquiry aid vs. a passive information tool (Avsheniuk et al., 2024). While research on AI in education continues to grow, the particular challenges faced by university ESP teachers remain under-explored. The literature has insufficiently explored ESP teachers' expectations and concerns for the impending influence of AI on language teaching, revealing a gap in ESP teachers' perceptions of AI in pedagogical initiatives.

Methods

The research study employed a qualitative approach to investigate the impact of AI tools on ESP teaching and learning. A structured questionnaire was developed based on a comprehensive review of relevant literature with three main dimensions: (1) academic integrity concerns, (2) overreliance on AI that hinders learning outcomes, and (3) challenges of integrating

AI into ESP teaching and learning. The data from the open-ended responses were analyzed using thematic analysis. The thematic analysis involved coding the responses to identify recurring themes and patterns related to the study's focus areas. The analysis was conducted using the reliability of coding assessed using the Miles-Huberman formula (Miles & Huberman, 2002). In addition to the questionnaire, classroom observations and literature analysis were incorporated to supplement the findings.

Participants

The research was carried out in a Ukrainian university with three faculties that teach ESP English for Specific Purposes for first- and second-year students. The study involved 16 ESP university teachers. The teachers were selected through purposive sampling to ensure they had previous experience working with AI in their teaching practice. Data was collected over the spring semester from March through June 2024. This sampling strategy enabled the researchers to survey ESP university teachers with diverse expertise in AI tools and language teaching, providing relevant qualitative insights that would be pertinent and meaningful for the qualitative data analysis. No translation was required since the survey was conducted in English.

Research instruments

The study employed multiple research instruments to investigate the impact of AI tools on ESP teaching and learning. A structured questionnaire was developed based on a comprehensive literature review, addressing three key dimensions: academic integrity concerns, overreliance on AI that hinders learning outcomes, and challenges in integrating AI into ESP instruction. The questionnaire included open-ended questions, allowing participants to provide detailed responses. Thematic analysis was used to analyze the responses, with coding conducted to identify recurring themes and patterns, ensuring reliability through the Miles-Huberman formula (Miles & Huberman, 2002). Classroom observations were conducted to supplement the questionnaire data to provide contextual insights into how AI tools were integrated into teaching practices and how students interacted with AI-generated content. A literature analysis was incorporated to support the study's findings and offer a broader perspective on the role of AI in ESP education.

Findings

Table 1. *Participant feedback on the integration of AI in ESP learning*

Theme	Frequency	Description
Overreliance on AI	12 (75%)	- Students depend heavily on AI to complete language tasks, particularly vocabulary, grammar, and case studies.
		- This overreliance reduces independent learning, problem-solving, and the development of critical thinking skills.
Skill Degradation	9 (56%)	- Continuous use of AI tools leads to a decline in language proficiency, especially in speaking and writing.
		-The heavy use of AI in case studies and tasks reduces the need for independent analysis and reflection.

Academic Integrity Concerns	10 (62%)	- Students often submit AI-generated content without adequate personal input or analysis.
		- AI-generated responses raise concerns about plagiarism, especially in tasks like vocabulary, grammar, and case studies.
Reduced Engagement in Learning	13 (81%)	- Overuse of AI tools diminishes student participation in tasks requiring critical thinking.
		- AI completion of case studies and tasks decreases active student involvement in the learning process.
Limited Speaking and Listening Skills	8 (50%)	- AI tools mainly focus on writing and grammar tasks.
		- The tools neglect speaking and listening practice, hindering holistic language development.
Contextual Understanding Issues	7 (44%)	- AI-generated content often lacks appropriate context, such as idiomatic expressions and cultural nuances.
		- AI struggles with capturing the subtleties required for meaningful language use, particularly in case studies.
Vocabulary Acquisition	11 (68%)	- Overuse of AI tools in completing vocabulary tasks may hinder natural word acquisition.
		- AI aids vocabulary learning but limits students' ability to internalize new words in context.
Use of AI in Completing Case Studies	14 (88%)	- AI tools are frequently used to complete case studies, leading to a reliance on AI for analysis and answers.
		- This reduces students' engagement in critical thinking and problem-solving, as they lean on AI-generated solutions.
Overuse of AI during Group Discussions	10 (62%)	- Students overuse AI in group discussions, relying on it to formulate ideas and responses.
		- This limits genuine peer interaction and impedes the development of communication and collaborative skills.

The qualitative data analysis from the participants' responses on using AI tools in their ESP courses identified several key themes that illustrate several opportunities for and obstacles to AI integration in language teaching. ESP university teachers were worried that students excessively rely on AI to end up with language assignments, such as vocabulary, grammar, and case studies. Most ESP university teachers reported this overdependence, which resulted in a noticeable drop in independent problem-solving, critical thinking, and analytical skills. Specifically, students use these AI tools to receive answers instantaneously, allowing them to skip the deeper cognitive processing that typically facilitates language learning. Besides, ESP teachers face a drop in global language competence, writing and speaking in particular, because of the regular use of AI. ESP university teachers (56%) reported that while AI tools helped with grammar correction and vocabulary enhancement, they diminished students' ability to produce language independently and spontaneously. The use of AI engines to generate content for case studies or assignments limited students' engagement with reflective learning, and thus, they failed to develop their language holistically. A common theme (62%) was academic integrity. ESP university teachers frequently reported limited analysis or personal input from students who submitted AI-generated content, raising concerns about plagiarism. Other teachers bemoaned the

automation of task completion — especially in vocabulary and grammar exercises — as a denigration of meaningful student work. The recent surge in this domain implies a space for the progressive ethics of using AI in language learning. AI tools were also found to deter students' engagement negatively. Students' active engagement in language learning tasks was reduced due to the overuse of AI, according to 81% of ESP university teachers. Since AI tools, just like other technology tools, provide ready-made solutions for tasks, students are less likely to think critically and discuss and collaborate. Such periods lead to the 'faulty' process of language acquisition and learning coupled with the inactivity on their part. Another concern (50%) is that AI tools primarily emphasize written language tasks — correcting grammar and enhancing vocabulary — while largely ignoring speaking and listening practice. According to relatively high ESP academic setting teachers, AI tools are useful for written tasks rather than being meaningful in providing opportunities for students to develop oral communication skills (found to be significant for overall language development in an ESP environment). 44% of ESP university teachers indicated that AI-produced content frequently contained insufficient context, especially idiomatic phrases or cultural background. The problem was especially relevant in case studies, where the AI was unable to understand the nuances of professional language or specific cultural environments, meaning it could not produce content that was accurate or relevant to students. Teachers highlighted that student need to be able to understand and use language within real-world contexts, and AI cannot fully replicate this experience. AI tools were found to aid vocabulary acquisition (68%). Professors mentioned that overdependence on AI for vocabulary activities could prevent students from internalizing new words in their context. Although AI can help when it comes to learning new words, it does not necessarily encourage learners to connect with vocabulary in actual contexts, thus limiting opportunities for contextual learning and retention. 88% of ESP university teachers responded that it was common practice to have students complete case studies using AI tools, which has resulted in an overdependence on AI to conduct analysis and solve problems. Because of this dependence, students avoided engaging with the case study process as they invariably seek AI for ready-made answers instead of possessing their ideas and critical thinking. ESP university teachers add that this robs students of the ability to work on real-life problems independently. Finally, 62% of ESP university teachers noticed that students resorted to using AI excessively when discussing in groups for ideas and structuring their answers. This reduced authentic peer engagement and collaborative learning since students often did not post their thoughts and ideas but used AI-generated suggestions. In-house communication suffered, preventing the building of social skills critical to academic and career success. The qualitative analysis suggests that although AI tools benefit ESP learning, there are several issues to resolve. An aspect of their use that ESP university teachers are apprehensive about is the overdependence on AI, which impedes autonomous learning, analysis-making, and developing speaking and listening skills.

ESP teachers have provided several comments about integrating AI tools into classroom activities. Reflect on the benefits and challenges of using AI for language learning tasks,

highlighting how students use these tools to complete assignments and engage with course material.

Teacher 1. *"As a teacher assessing the writing task, I discovered that 7 of the 15 students had used AI tools – like ChatGPT – to perform the letter of complaint assignment. Although the letters were neatly formatted, there was a clear disconnect in how these students approached the task. The AI-generated a very polished and professional letter, but the students lost the opportunity to develop critical thinking, problem-solving and writing skills. Such overreliance on AI raises concerns about academic integrity, as it diminishes the significance of independent effort and original work. Additionally, it hinders students from expressing their thoughts and reflecting on the writing process. You want them to be actively engaged in doing these tasks independently and learning those skills."*

Teacher 2. *"Although I have just recently started the students' in-home assignments on vocabulary, I discovered that many of them regularly used AI tools like ChatGPT to do assignments on gap-filling tasks, writing on words and phrases together and matching exercises. However, though the AI answers were accurate, and hereafter, being satisfied with the assignment, it came through clearly to us that the students were leaning too much on the tool to solve these tasks. However, this heavy dependence casts doubt on their involvement in learning, restricting their capacity to assimilate and remember the vocabulary autonomously. None of the students had to think critically about what the words meant or how they fit into longer sentences. Once they had the words, they could ask the program for an example sentence, further detracting from their chances of reinforcing their new vocabulary."*

Teacher 3. *"When working in groups on case studies, it became apparent that quite a few students were using AI tools to generate ideas, especially with complex vocabulary that was clearly beyond their level of proficiency. The sophisticated language used in their answers was readily recognized as beyond their usual level of language competencies. When asked to explain the meaning of this advanced vocabulary, most students could not provide an understanding of the term or context in which it was used, suggesting that they did not fully grasp the word being used. It showcases a disturbing phenomenon where AI assists students to sound more intelligent than they are, and in doing so, they mute their learning. Relying on AI to generate ideas and vocabulary leaves them without the critical thinking and problem-solving skills they need, and they fall short of learning new language skills meaningfully. This highlights the importance of showing value in how the students respond during case study discussions and using language that fits where they are now and how they think."*

Teacher 4. *"As part of a reading task, which asked students to determine true and false statements about a reading passage, I noted that several of these students appeared to be completing the task using AI. They answered with excessive confidence, sometimes selecting answers that were too simple or irrelevant to the passage. Upon reviewing the responses, I recognized that some of my students relied on AI to create reasonable but broadly framed responses without the depth of connection to the texts they were reading. When asked to justify*

their reasoning, the students struggled to explain their decisions in detail, suggesting that they had relied on AI responses without understanding the material. However, this is yet another indication that although AI is able to provide instant answers, it allows no room for comprehension of the content, as students are losing out on the analysis and reasoning required to perform such tasks. Instead of relying on AI to provide ready-made answers, students need to be directed toward developing their analysis and interpretation of reading materials."

Teacher comments reflect emerging warnings regarding the overuse of AI applications in ESP classes, especially concerning student engagement, critical thinking, and vocabulary acquisition. As AI tools become more integrated into the classroom, educators need to balance utilizing the technology efficiently with encouraging independent learning and authentic engagement with the language.

As observed, overreliance on AI tools significantly influences final test results and overall learning outcomes. Teachers noted that while AI tools helped students to complete tasks with apparent ease – such as filling in vocabulary gaps, writing assignments, or case studies – their performance often lacked depth and critical engagement with the material. In particular, students who consistently used AI to complete writing tasks or vocabulary assignments could not demonstrate the same level of language proficiency when assessed in more complex, open-ended contexts. Teachers observed that while AI-generated responses for tasks like writing letters of complaint or completing vocabulary exercises were technically correct, students struggled with creating original, contextually rich responses during assessments. This became evident in final tests, where students were asked to apply their knowledge in unfamiliar or more challenging contexts. Many students who had relied heavily on AI for task completion failed to connect vocabulary or phrases with their understanding, leading to superficial responses that lacked personal insight or coherence.

Using AI in group discussions and case studies also limited students' ability to engage meaningfully with the material. Teachers noted that during final assessments, students who had overused AI to generate complex vocabulary in case study reports could not correctly explain or use the vocabulary in real-time scenarios. This lack of comprehension led to lower overall test scores, particularly in tasks that required deeper analysis and critical thinking.

Moreover, students who frequently used AI to complete vocabulary and grammar tasks often showed gaps in their ability to apply new words or structures in context, which became evident in the final tests. While AI helped them complete exercises, it did not foster long-term retention or the ability to use language flexibly in real-world situations. As a result, these students performed below expectations in assessments that required them to demonstrate a more holistic understanding of the language.

The reliance on AI for quick fixes and pre-generated responses reduced students' ability to engage in independent learning and critical reflection. This, in turn, led to lower performance in final assessments, where the ability to apply language skills in real-world contexts is essential.

Therefore, teachers emphasized the need for a balanced approach, where AI tools support learning but do not replace the essential cognitive processes involved in language acquisition.

Discussion

The infusion of AI in education is a multifaceted and dynamic development that has profound implications for universities. AI is here to stay in education, and its role in education is expected to grow substantially in the near future (Miao & Holmes, 2023). Although AI can provide high efficiency, personalized learning, innovative teaching methods, and other benefits, it could also lead to some critical risks and challenges that need comprehensive ethical reflection. Overreliance on AI is one of the essential things found from the analysis. ESP university teachers have noted it is increasingly common for students to rely on AI to complete language tasks, from working on vocabulary exercises to writing essays, and even completing case-study analyses. This dependence had caused disengagement of students on basic cognitive steps of the language process involving problem-solving, critical analysis, and self-reflection. In particular, tasks once involving independent thinking – writing, letters of complaint, case studies, whatever – are canned now, often by AI, making it difficult to internalize and apply the knowledge gained. Excessive overreliance on AI, having an AI chatbot do all the work instead of allowing students to understand the material in depth. As needed as it may be, AI has its limitations: the core cognitive processes of ongoing practice and acquisition will always be best lead by a human brain rather than a click of a button. This includes the skill degradation seen among students, especially in speaking and writing. Although they can assist with grammar corrections and vocabulary improvements, AI tools do not offer opportunities for students to practice what their spoken language sounds like or use language creatively in the world outside of classes. Both the generation of case study responses and the writing up of complaint letters lend themselves to routine overuse of AI (popping in details from an assignment, calling for a case study response, having the system generate a solution, which if successful is simply replicated by the student rather than being reconstructed in an original new response). As a result, students often fail to develop their language production skills, which are vital for successful communication in professional ESP contexts. ESP teachers noted that this deficit in language production skills becomes particularly apparent in speaking and listening exercises, where AI tools are limited in their capacity to provide meaningful interactive feedback or opportunities for authentic language use. It supports the foremost scholar's concern regarding AI-assisted language learning, which is the risk of overreliance on the tool.

ESP teachers report that students often submit AI-generated content with minimal personal analysis or modification, raising concerns about plagiarism and dishonest academic practices. According to Alshater (2023), AI tools such as ChatGPT allow students to produce content without working through the writing process or producing new analyses. This challenge is especially pertinent in ESP courses where linguistic competence and the ability to use professional language are of the utmost concern. Kostka and Toncelli (2023) support this by stating that using

AI-generated content allows students to "skip" critical thinking, resulting in work with little academic work put into it. Guerrero-Dib et al. (2020) and McCabe et al. (2001) further argue that AI tools make academic misconduct, including plagiarism, more convenient for students and allow them to turn in work that does not reflect their intellectual contributions. As Uzun and Kilis (2020) also point out, the ease of access to these tools often leads students to submit work without proper effort. ESP university professors highlight the need for clear ethical guidelines regarding AI usage. They use it as an enhancement rather than a routine substitute for one of the most precious components of every person's learning journey and a developing critical thinking process. Educators can mitigate the effects of AI misuse in ESP courses, using practical insights by aligning their ESP practices with these insights to balance generating AI's advantages and promoting academic integrity. Academics stress its multidimensional character, such as reasoning, inference, judgement (critical), metacognition, analysis and creativity (Facione, 2013). One of the main problems found was that students who overly rely on AI tools appear less engaged cognitively. Active learning tasks, like group discussions and problem-solving exercises, are diminishing as students turn to AIs for quick solutions. Moreover, this dependence may restrict collaborative and reflective processes necessary for promoting critical thinking and finally leading to successful language learning in ESP settings.

This study's findings reveal significant parallels with scholarly discussions, particularly in the role of AI tools like ChatGPT in shaping critical thinking and related skills. As noted by Pang (2022), Trang (2021), and Young (2003), AI tools hold promise for enhancing learners' critical thinking, media literacy, and digital navigation skills. However, the study also highlights challenges associated with AI usage that can detract from these goals.

The excessive use of AI tools has diminished creativity and the subject matter grasp, especially where the contextual application of language skills is required. In addition, in case studies and group work tasks, students use AI to generate advanced vocabulary and construct structured responses without understanding or applying the material within context. Such an approach stifles critique and genuine engagement, resulting in lower-quality student work and missed opportunities for authentic learning – a key problem in ESP courses, where students are expected to address real-life problems and use the language in a professional context. Mohammad et al. (2023) further highlighted the speed at which ChatGPT can produce inaccurate information and biases, its ability to discourage critical thinking, and ethical issues related to cheating and privacy. Notwithstanding these challenges, the benefits of leveraging AI tools in ESP courses are considerable. In addition, AI can promote autonomic learning and individualized learning by offering personalized feedback and resources when rightly integrated. This allows students to work alone and repeat exercises to gain a deeper understanding. According to some ESP university teachers, AI tools are particularly effective when used as a supplemental rather than a primary learning aid, supporting vocabulary acquisition and grammar improvement. The use of artificial intelligence in educational environments is disruptive to classical pedagogical paradigms and has

important implications for higher education institutions. AI is already highly attracted to education, and its contribution is predicted to grow in the coming years (Miao & Holmes, 2023).

ESP university teachers noticed that a lack of language production skills is most perceivable in speaking and listening activities, where AI tools have limited potential to offer meaningful interactive feedback or authentic language use opportunities. It attests to the first scholar's worry about AI-facilitated language learning, which is the potential for dependence on the tool. ESP university teachers find that students frequently turn in minuscule AI-generated content accompanied by personal analysis or modification, thus striking the fear of plagiarism and dishonest academia. The issue identified by ESP university teachers is in line with the evidence documented in the literature. As Alshater (2023) notes, AI tools such as ChatGPT allow students to generate content without either going through the writing process or providing original analysis. This concern is especially pertinent in ESP courses in which linguistic competence and application of the language pertinent to the profession are central. As Kostka and Toncelli (2023) put it, AI-generated content helps students skip the critical thinking step necessary to make their work seem like it was produced through academic effort. Guerrero-Dib et al. (2020) and McCabe et al. (2001) argue that the nature of AI tools presents the risk of academic dishonesty and plagiarism, allowing students to submit work that does not represent the contribution of their intellect(s). According to Uzun and Kilis, these tools are easily accessible, so students submit with little to no effort. In turn, ESP university instructors emphasize the need for explicit guidelines on the ethical dimensions of AI and use so that technologies serve to supplement, rather than replace, rich engagement with learning. Such alignment of practices with insights can assist educators in approaching the positive impacts of AI while ensuring academic integrity in ESP courses. Scholars highlight its complex nature – reasoning, inference, critical judgment and creative thinking (Facione, 2013). According to Facione, critical thinking is the purposeful interpretation, analysis, evaluation, and inference based on evidence and context. The renewed attention on students' reasoning ability ties into the ongoing conversation about cultivating critical thinking skills. One concern is the lack of cognitive engagement for students who depend excessively on AI tools. The increased reliance on AI for quick answers negatively impacts tasks that require students to actively engage in their learning, like group discussions and problem-solving work. This dependence restricts the collaborative and reflective processes necessary for cultivating thinking skills and attaining successful EAP courses in language learning. While AI tools can efficiently handle simple tasks, they have also been associated with a decline in creativity and shallow comprehension of relevant material, especially in tasks that require the application of language skills in context. Moreover, students copy and paste AI content to build up complex vocabulary or create structured answers in activities such as case studies and group work despite not understanding the content they use. This threatens critical approaches and moves away from real engagement, but importantly, there are lost potentials for quality output and learning movements – especially in ESP-type courses; both worlds of applied work and language of the real world lie

within its tent, where real-world issues, challenges and professional language usage are means and ends.

Mohammad et al. (2023) highlighted the inaccuracy and bias of ChatGPT, as well as its potential to discourage critical thinking, as well as ethical (e.g. cheating) and privacy concerns. Notwithstanding these challenges, the potential advantages of using AI tools in ESP instruction are considerable. AI can support learner independence and individualized learning through adaptive feedback when employed responsibly. That allows students to move at their own pace and return to previous tasks for deeper understanding. When incorporated as a supplementary tool rather than a substitute for active learning practices, ESP university teachers have observed that AI tools can effectively support literacy for vocabulary acquisition and grammar increase. With that said, this finding highlights the need to keep a fine balance on the usage of AI. Educators will need to teach students to use AI tools responsibly, supplementing rather than supplanting critical thinking and engagement. However, applying these elements of AI as pacemakers of active learning, even beyond the ESP pedagogic commitment, can greatly benefit students in acquiring the specialized academic and professional language needed for their future roles in their ESP courses. Overcoming these challenges will require a collaborative effort from all relevant stakeholders in the ELT community (including educators, policymakers, technology developers, and researchers) to establish ethical frameworks, effective pedagogical approaches, and supportive policies for the integration of AI into language teaching importance of achieving a balance in AI usage.

Conclusion

The use of AI tools in ESP courses has revealed promising opportunities and considerable challenges (Beckman et al., 2023). The study uncovers that despite the benefits of AI tools like ChatGPT in areas such as vocabulary and grammar improvement, as well as case studies, the students' overreliance on these tools has significant implications for their language learning, engagement and academic honesty. Overreliance on AI results in less independent thought and less working on tasks where you need to apply critical thought or engage with language in the real world, especially in speaking and listening-related tasks. Another issue includes academic integrity, as many students submit AI-generated content without regional study, often leading to plagiarism and reducing originality in their academic work. Students' understanding of vocabulary in context and case studies – contributing to language use in ESP contexts and developing their influence with language in their professional lives – can significantly decrease with excessive AI production. Nonetheless, if used appropriately, AI tools can help students learn at their own pace and expand their vocabulary, among other benefits. They may serve as an additional resource to cement and reiterate knowledge and allow learners to learn at their own pace. But the secret to their power lies in striking the right balance, leveraging AI to assist with language learning and keeping students engaged with their education.

Based on the data from the surveys carried out among ESP university teachers, specific recommendations were provided to tackle problems arising from AI technology in the ESP teaching scope. Teachers stressed that maintaining an active dialogue with students on vocabulary tasks would build dialogue for language and thinking across students and build those skills rather than simply trusting in the structure generated by AI. In addition, they encourage the promotion of authentic language used in the case study discussions by stating that "It is important to scaffold students to use language in a way that makes sense for their level." Moreover, teachers stressed the importance of analytical skills for students, which means reading and analyzing books independently rather than relying upon AI to produce ready-made answers. Among the key recommendations was establishing clear guidelines for using AI and ensuring that AI is positioned as an academic aide, not an academic substitute. Teachers also recommended incorporating AI tools into structured activities where students become critical consumers of AI-generated content instead of accepting it at face value. Therefore, when it comes to the balance of technology tools, it's important to regularly reflect on the strengths and weaknesses of these AI tools and encourage teacher-student cooperation to ensure that students remain closely linked and receive direction from you to mix AI with their individual strategies for learning. Educators should additionally foster ethical AI usage by teaching students about academic integrity, data privacy, and the tendency of AI-generated content to be biased. It is important to lay out what appropriate AI behaviour looks like, and this includes expectations around properly citing the use of AI in specific academic work, or else academic cheating will become rampant.

The sampling merit of this study is in the limited sample of ESP university teachers and students, a small number of ESP university teachers and students (participants), which we realize is not such a representative segment of the population of ESP learners in general, and this could centre on a limitation of the findings of this study, i.e. the generalizability of (search) findings. The feedback collected was primarily focused on ESP university teachers. It lacked input from students, whose perspective could lead to a fuller understanding of how AI tools shape their learning. Another avenue for potential bias is that they were self-reported data from ESP university teachers, whose concept of AI potential influence could not necessarily align with their student's behaviours or learning performance, thus contributing to misperceptions. Despite the limitations of this study, which only considers generalization in the ESP context, it makes a good case for future research on integrating AI into ESPs easily to develop more effective teaching.

Funding: This research is not funded.

Acknowledgments: Not applicable.

Conflicts of Interest: The authors declare no conflict of interest.

Authenticity: This manuscript is an original work.

Artificial Intelligence Statement: During the preparation of this work the authors used DeepL Translate, Smodin, Grammarly, QuillBot in order to improve readability and language. After using

this tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

About the Authors

Dr. Nataliia Avsheniuk is a distinguished scholar in education, a Professor and a Doctor of Science in Education. She leads the Foreign Systems of Pedagogical and Adult Education Department at the Ivan Ziaziun Institute of Pedagogical and Adult Education, which is part of the National Academy of Educational Sciences of Ukraine. She is recognized for her significant contributions to educational research and policy in Ukraine. Email: nataliya.avshenyuk@gmail.com ORCID ID: <https://orcid.org/0000-0003-1012-005X>

Dr. Nataliia Seminikhyna, Assistant Professor in the Department of Foreign Languages at the Faculty of Economics, Taras Shevchenko National University of Kyiv. Her work focuses on pedagogical methods in language education, particularly in the context of economic studies. ORCID ID: <https://orcid.org/0000-0001-6246-4132>

Dr. Larysa Ruban, Associate Professor, Department of Foreign Languages, Faculty of Economics, Taras Shevchenko National University of Kyiv. She holds a PhD in Education and has contributed to various publications in the field of education and ESP teaching. Her expertise lies in the intersection of language studies and economics, providing students with the essential language skills required in the business and economic environment. Email: larysa.ruban@knu.ua ORCID ID: <https://orcid.org/0000-0001-5539-931X>

Dr. Yuliia Sviatiuk, Associate Professor, Department of Foreign Languages, Taras Shevchenko National University of Kyiv. She holds a PhD in Philology and her research interests include cognitive linguistics, applied linguistics, cross-cultural communication, and translation studies. Email: julia.sviatiuk@knu.ua ORCID ID: <https://orcid.org/0000-0001-8343-1890>

References

- Abdelghani, R., Wang, Y., Yuan, X., Bouzid, Y., & Liu, X. (2024). GPT-3-driven pedagogical agents to train children's curious question-asking skills. *International Journal of Artificial Intelligence in Education*, 34, 483–518. <https://doi.org/10.1007/s40593-023-00340-7>
- Akbar, L., & Thomson, D. (2016). Academic integrity: The fool's dilemma. *International Journal of Digital Society*, 7(5), 1165–1170.
- Alshater, M. (2023). Exploring the Role of Artificial Intelligence in Enhancing Academic Performance: A Case Study of ChatGPT. Available at SSRN: <https://ssrn.com/abstract=4312358> or <http://dx.doi.org/10.2139/ssrn.4312358>.

- Avsheniuk, N., Lutsenko, O., Svyrydiuk, T., & Seminikhyna, N. (2024). Empowering Language Learners' Critical Thinking: Evaluating ChatGPT's Role in English Course Implementation. *Arab World English Journal (AWEJ) Special Issue on ChatGPT*, April 2024: 210-224. DOI: <https://dx.doi.org/10.24093/awej/ChatGPT.14>
- Chaudhry, M., Cukurova, M., & Luckin, R. (2022). A transparency index framework for AI in education. *EdArXiv Preprints*. <https://doi.org/10.35542/osf.io/bstcf>
- Cotton, D., Cotton, P., & Shipway, J. (2023). Chatting and cheating: Ensuring academic integrity in the era of ChatGPT. *Innovations in Education and Teaching International*, 1-12. <https://doi.org/10.1080/14703297.2023.2190148>
- Eaton, S., & Edino, R. (2018). Strengthening the research agenda of educational integrity in Canada: A review of the research literature and call to action. *International Journal of Educational Integrity*, 14(1).
- Facione, P. (2015). *Critical Thinking: What it Is and Why it Counts*. Millbrae, CA: California Academic Press.
- Guerrero-Dib, J., Derbez, L., & Heredia, Y. (2020). Impact of academic integrity on workplace ethical behaviour. *International Journal for Educational Integrity*, 16. <https://doi.org/10.1007/s40979-020-0051-3>
- Hanim, N., Aripin, N., & Lin, N. (2020). Exploring the connection between critical thinking skills and academic writing. *International Journal of Asian Social Science*, 10(2), 118–128. <https://doi.org/10.18488/journal.1.2020.102.118.128>
- Huberman, A., & Miles, M. (2002). *The qualitative researcher's companion*. <https://doi.org/10.4135/9781412986274>
- Kaepfel, K. (2021). The influence of collaborative argument mapping on college students' critical thinking about contentious arguments. *Thinking Skills and Creativity*, 40, 100809. <https://doi.org/10.1016/j.tsc.2021.100809>
- Kasneci, E., Sessler, K., Küchemann, S., Bannert, M., Dementieva, D., Fischer, F., Gasser, U., Groh, G., Günemann, S., Hüllermeier, E., Krusche, S., Kutyniok, G., Michaeli, T., Nerdel, C., Pfeffer, J., Poquet, O., Sailer, M., Schmidt, A., Seidel, T., Stadler, M., Weller, J., Kuhn, J., Kasneci, G. (2023). ChatGPT for good? On opportunities and challenges of large language models for education. *Learning and Individual Differences*, 103, 102274. <https://doi.org/10.1016/j.lindif.2023.102274>
- Kem, D. (2022). Personalised and adaptive learning: Emerging learning platforms in the era of digital and smart learning. *International Journal of Social Science and Human Research*, 5(2), 385–391.
- Kostka, I., & Toncelli, R. (2023). Exploring applications of ChatGPT to English language teaching: Opportunities, challenges, and recommendations. *On the Internet*, 27(3). Northeastern University, USA. <https://doi.org/10.55593/ej.27107int>
- Kovačić, A., & Bubaš, G. (2024). Using conversational artificial intelligence for online learning activities in English for specific purposes: A pilot study of students' experiences with

- Bing Chat. In E. Delle Donne (Ed.), *Proceedings of the 16th International Conference Innovation in Language Learning* (pp. 1–5). Filodiritto Publisher.
- Liu, C., Hou, J., Tu, Y.-F., Wang, Y., & Hwang, G.-J. (2023). Incorporating a reflective thinking promoting mechanism into artificial intelligence-supported English writing environments. *Interactive Learning Environments*, 31(9), 5614–5632. <https://doi.org/10.1080/10494820.2021.2012812>
- Mahmood, Z. (2009). Students' understanding of plagiarism and collusion and recommendations for academics. *WSEAS Transactions on Information Science and Applications*, 6, 1349–1358. <https://api.semanticscholar.org/CorpusID:55373192>
- Marengo, A., Pagano, A., Pange, J., & Soomro, K. A. (2024). The educational value of artificial intelligence in higher education: A 10-year systematic literature review. *Interactive Technology and Smart Education*, 21(4), 625–644. <https://doi.org/10.1108/ITSE-11-2023-0218>
- Maurer, H., Kappe, F., & Zaka, B. (2006). Plagiarism – A survey. *Journal of Universal Computer Science*, 12(8), 1050–1084.
- McCabe, D., Trevino, L., & Butterfield, K. (2001). Cheating in academic institutions: A decade of research. *Ethics & Behavior*, 11(3), 219–232. https://doi.org/10.1207/S15327019EB1103_2
- Meuschke, N., & Gipp, B. (2013). State-of-the-art in detecting academic plagiarism. *International Journal for Educational Integrity*, 9, 50–71. <https://doi.org/10.5281/zenodo.3482941>
- Miao, F., & Holmes, W. (2023). *Guidance for generative AI in education and research*.
- Mohammad, B., Supti, T., Alzubaidi, M., Shah, H., Alam, T., Shah, Z., & Househ, M. (2023). The pros and cons of using ChatGPT in medical education: A scoping review. *Studies in Health Technology and Informatics*, 305, 644–647. <https://doi.org/10.3233/SHTI230580>
- Ningsih, N. (2023). The importance of game-based learning in English learning for young learners in the 21st century. *The Art of Teaching English as a Foreign Language*, 4, 25–30.
- Park, C. (2004). Rebels without a clause: Towards an institutional framework for dealing with plagiarism by students. *Journal of Further and Higher Education*, 28(3), 471–488. <https://doi.org/10.1080/0309877042000241760>
- Qasem, H., Asaad, M., & Shabdin, A. A. (2021). The predictive role of morphological awareness and productive vocabulary knowledge in L2 postgraduate students' academic writing. 7(1), 24–44.
- Stoesz, B., & Eaton, S. (2020). Understanding the academic integrity policies of publicly funded universities in western Canada. *Educational Policy*. <https://doi.org/10.1177/0895904820983032>

- Trang, T. (2021). Students' Evaluation of the Outcomes Achieved from Their Learning Experience at Industrial University of Ho Chi Minh City. *Journal of Science and Technology – IUH*, 53(5). <https://doi.org/10.46242/JSTIUH.V53I05.4146>
- Uzun, A., & Kilis, S. (2020). Investigating antecedents of plagiarism using the extended theory of planned behavior. *Computers & Education*, 144, 103700. <https://doi.org/10.1016/j.compedu.2019.103700>
- Young, S. (2003). Integrating ICT into second language education in a vocational high school. *Journal of Computer Assisted Learning*, 19(4), 447–461. <https://doi.org/10.1046/J.0266-4909.2003.00049.X>